Exercise 5 – Reactive Forms

Install Reactive Forms

```
npm install
```

---

app.config.ts

```
import { ApplicationConfig } from '@angular/core';
import { provideRouter } from '@angular/router';
import { provideForms, provideReactiveForms } from '@angular/forms';
import { routes } from './app.routes';

export const appConfig: ApplicationConfig = {
 providers: [
  provideRouter(routes),
  provideForms(),
  provideReactiveForms()
 ]
};
```

Generate Component

```
ng generate component pages/reactive-enrollment
```

---

app.routes.ts

```
import { Routes } from '@angular/router';
import { HomeComponent } from './pages/home/home.component';
import { CourseListComponent } from './pages/course-list/course-list.component';
import { StudentProfileComponent } from './pages/student-profile/student-profile.component';
```

```
import { EnrollmentFormComponent } from './pages/enrollment-form/enrollment-form.component';

import { ReactiveEnrollmentComponent } from './pages/reactive-enrollment/reactive-enrollment.component';


export const routes: Routes = [

{path:'',component:HomeComponent},

{path:'courses',component:CourseListComponent},

{path:'profile',component:StudentProfileComponent},

{path:'enroll',component:EnrollmentFormComponent},

{path:'reactive',component:ReactiveEnrollmentComponent}

];
```

---

reactive-enrollment.component.ts

```
import { Component } from '@angular/core';

import { FormArray, FormBuilder, ReactiveFormsModule, Validators } from '@angular/forms';


@Component({

selector:'app-reactive-enrollment',

standalone:true,

imports:[ReactiveFormsModule],

templateUrl:'./reactive-enrollment.component.html',

styleUrls:['./reactive-enrollment.component.css']

})

export class ReactiveEnrollmentComponent{


constructor(private fb:FormBuilder){}
```

```
enrollmentForm=this.fb.group({
name:['',Validators.required],
email:['',[Validators.required,Validators.email]],
mobile:['',[Validators.required,Validators.pattern('[0-9]{10}')]],
branch:['',Validators.required],
semester:['',Validators.required],
skills:this.fb.array([
this.fb.control('',Validators.required)
])
});

get skills(){
return this.enrollmentForm.get('skills') as FormArray;
}

addSkill(){
this.skills.push(this.fb.control('',Validators.required));
}

removeSkill(index:number){
this.skills.removeAt(index);
}

submit(){
console.log(this.enrollmentForm.value);
this.enrollmentForm.markAllAsTouched();
}
```

```
reset(){
this.enrollmentForm.reset();
this.skills.clear();
this.skills.push(this.fb.control('',Validators.required));
}

}
```

---

reactive-enrollment.component.html

```
<h2>Reactive Enrollment Form</h2>

<form [formGroup]="enrollmentForm" (ngSubmit)="submit()">

<label>Name</label>
<input type="text" formControlName="name">

<div
*ngIf="enrollmentForm.get('name')?.invalid&&enrollmentForm.get('name')?.touched">
Name Required
</div>

<label>Email</label>
<input type="email" formControlName="email">

<div
*ngIf="enrollmentForm.get('email')?.invalid&&enrollmentForm.get('email')?.touched">
Valid Email Required
</div>
```

```
<label>Mobile</label>

<input type="text" formControlName="mobile">


<div
*ngIf="enrollmentForm.get('mobile')?.invalid&&enrollmentForm.get('mobile')?.touched
">

Enter 10 Digit Mobile Number

</div>


<label>Branch</label>

<input type="text" formControlName="branch">


<label>Semester</label>

<input type="number" formControlName="semester">


<h3>Skills</h3>


<div formArrayName="skills">

<div *ngFor="let skill of skills.controls;let i=index">

<input [formControlName]="i">

<button type="button" (click)="removeSkill(i)">Remove</button>

</div>

</div>


<button type="button" (click)="addSkill()">Add Skill</button>


<br><br>


<button type="submit" [disabled]="enrollmentForm.invalid">Submit</button>
```

```
<button type="button" (click)="reset()">Reset</button>

</form>

<div *ngIf="enrollmentForm.valid">

<h3>Submitted Data</h3>

<p>Name : {{enrollmentForm.value.name}}</p>
<p>Email : {{enrollmentForm.value.email}}</p>
<p>Mobile : {{enrollmentForm.value.mobile}}</p>
<p>Branch : {{enrollmentForm.value.branch}}</p>
<p>Semester : {{enrollmentForm.value.semester}}</p>

<ul>
<li *ngFor="let skill of enrollmentForm.value.skills">
{{skill}}
</li>
</ul>

</div>
```

---

reactive-enrollment.component.css

```
form{
width:450px;
padding:20px;
border:1px solid #ccc;
```

```
border-radius:10px;
}
label{
display:block;
margin-top:15px;
font-weight:bold;
}
input{
width:100%;
padding:8px;
margin-top:5px;
}
button{
margin-top:10px;
margin-right:10px;
padding:8px 16px;
}
div{
margin-top:5px;
}
```

---

header.component.html

```
<header>
<h1>{{title}}</h1>
<nav>
<a routerLink="/">Home</a>
<a routerLink="/courses">Courses</a>
<a routerLink="/profile">Profile</a>
```

```
<a routerLink="/enroll">Template Form</a>

<a routerLink="/reactive">Reactive Form</a>

</nav>

</header>
```

Run

ng serve

Open:

http://localhost:4200/reactive

Output Screenshot: